

Daily Geometry Drill #2 — Answers & Investigations

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Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Circle Equations · Tangents · Line–Circle Tests · Chord Lengths.

Section A Rapid Recognition

1. Write down the centre of the circle $x^2 + y^2 - 6x - 8y + 24 = 0$.

Answer: "(3, 4)"

Method: Complete the square: $(x - 3)^2 + (y - 4)^2 = 25 - 24 = 1$, so centre (3, 4), radius 1.

Investigate further: In $x^2 + y^2 + 2gx + 2fy + c = 0$ the centre is $(-g, -f)$ and here $2g = -6$, $2f = -8$, so $g = -3$, $f = -4$ — centre (3, 4) ✓. **If the coefficients of x^2, y^2 are not 1, divide through first so they are, then read the centre.**

2. Write down the radius of the circle $x^2 + y^2 - 6x - 8y + 24 = 0$.

Answer: 1

Method: From A1, $(x - 3)^2 + (y - 4)^2 = 1$, so $r = \sqrt{1} = 1$.

Investigate further: Formula check: $r = \sqrt{g^2 + f^2 - c} = \sqrt{9 + 16 - 24} = \sqrt{1} = 1$ ✓. **A "circle" with $g^2 + f^2 - c < 0$? Empty locus — the square sum equals a negative number.**

3. Write down the equation of the circle with centre (2, -1) and radius 5.

Answer: " $x^2 + y^2 - 4x + 2y - 20 = 0$ "

Method: $(x - 2)^2 + (y + 1)^2 = 25$ expands to $x^2 + y^2 - 4x + 2y + 5 = 25$, i.e. $x^2 + y^2 - 4x + 2y - 20 = 0$.

Investigate further: Reading back: $2g = -4 \Rightarrow g = -2$ (centre $x = 2$), $2f = 2 \Rightarrow f = 1$ (centre $y = -1$), $r = \sqrt{g^2 + f^2 - c} = \sqrt{4 + 1 + 20} = 5$ ✓. **Rewrite for centre (2, -1) radius 2: what becomes of the constant? $x^2 + y^2 - 4x + 2y + 1 = 0$.**

4. Write down the radius of the circle $x^2 + y^2 - 4x + 6y - 12 = 0$.

Answer: 5

Method: Complete the square: $(x - 2)^2 + (y + 3)^2 = 4 + 9 + 12 = 25$, so $r = 5$.

Investigate further: Formula: $g = -2, f = 3, r = \sqrt{4 + 9 + 12} = 5$ ✓. **All these squares are perfect: $5^2, 4^2, 3^2$. What radius would an *irrational* answer look like, and would that stop a clean distance test? Only the appearance changes; the distance test still works.**

5. The point $(x, 4)$ lies on the circle $x^2 + y^2 = 25$ with $x > 0$. Find x .

Answer: 3

Method: $x^2 + 4^2 = 25 \Rightarrow x^2 = 9 \Rightarrow x = \pm 3$; take +3.

Investigate further: Both $(3, 4)$ and $(-3, 4)$ lie on the circle — the $x > 0$ condition kills one branch.

List all lattice points (x, y) **on** $x^2 + y^2 = 25$. $(0, \pm 5), (\pm 3, \pm 4), (\pm 4, \pm 3), (\pm 5, 0)$.

6. The radius from the origin to the point $(3, 4)$ on $x^2 + y^2 = 25$ has what gradient?

Answer: $\frac{4}{3}$

Method: Gradient = $\frac{4-0}{3-0} = \frac{4}{3}$.

Investigate further: The radius is the line joining centre to the point. **Gradient of the perpendicular radius to** $(3, -4)$? $-\frac{4}{3}$.

7. The tangent to $x^2 + y^2 = 25$ at the point $(3, 4)$ has what gradient?

Answer: $-\frac{3}{4}$

Method: Tangent $\perp$ radius, and radius gradient is $\frac{4}{3}$, so tangent gradient = $-\frac{1}{4/3} = -\frac{3}{4}$.

Investigate further: The tangent line is $3x + 4y = 25$ (via $xx_1 + yy_1 = r^2$). **Does it pass through** $(3, 4)$? $9 + 16 = 25$ ✓. **What is its x-intercept?** $x = 25/3$.

8. Write down the equation of the circle with centre $(-1, 2)$ and radius 3.

Answer: " $x^2 + y^2 + 2x - 4y - 4 = 0$ "

Method: $(x + 1)^2 + (y - 2)^2 = 9$ expands to $x^2 + y^2 + 2x - 4y + 5 = 9$, i.e. $x^2 + y^2 + 2x - 4y - 4 = 0$.

Investigate further: Check centre: $g = 1 \Rightarrow$ centre $x = -1$, $f = -2 \Rightarrow$ centre $y = 2$ ✓; $r = \sqrt{1 + 4 + 4} = 3$ ✓. **Where does this circle cross the axes?** Put $y = 0$: $x^2 + 2x - 4 = 0$, a surd — compare with the clean integer axes of Day 1 right-triangles.

9. The centre of $x^2 + y^2 - 6x - 8y + 24 = 0$ is $(3, 4)$. Find the perpendicular distance of this centre from the line $4x + 3y - 9 = 0$.

Answer: 3

Method: Distance = $\frac{|4(3) + 3(4) - 9|}{\sqrt{4^2 + 3^2}} = \frac{|12 + 12 - 9|}{5} = \frac{15}{5} = 3$.

Investigate further: This is day 1's perpendicular-distance formula applied to a circle's centre. **Since** $d = 3 > r = 1$, **the line misses the circle. How many common tangents could such an external line family have?** Preview of Day 3's tangent-counting.

10. A chord of the circle $x^2 + y^2 = 25$ is at perpendicular distance 3 from the origin. Find the length of the chord.

Answer: 8

Method: Half-chord = $\sqrt{r^2 - d^2} = \sqrt{25 - 9} = \sqrt{16} = 4$; full length = 8.

Investigate further: The perpendicular from the centre to a chord bisects it — the half-chord

formula. **If the distance were 4, the length is 6; if 5, it degenerates to a tangent. Check both.** The chord shrinks to a point exactly when $d = r$.

Section B Manipulation Drills

1. Which of the following is the equation of the circle with centre $(3, -2)$ and radius 4?

A) $x^2 + y^2 - 6x + 4y - 3 = 0$

B) $x^2 + y^2 - 6x + 4y + 9 = 0$

C) $x^2 + y^2 + 6x - 4y - 3 = 0$

D) $x^2 + y^2 - 6x - 4y - 3 = 0$

Answer: A) $x^2 + y^2 - 6x + 4y - 3 = 0$

Method: $(x - 3)^2 + (y + 2)^2 = 16$ gives $x^2 + y^2 - 6x + 4y + 13 = 16$, i.e. $x^2 + y^2 - 6x + 4y - 3 = 0$.

Investigate further: Only A has both the x -coefficient -6 (centre $x = 3$) and the y -coefficient $+4$ (centre $y = -2$). **B has the right coefficients but the wrong constant $+9$; what radius would that give? $r = \sqrt{9 + 4} = 5$.**

2. The circle $x^2 + y^2 + 2x - 4y - 11 = 0$ has what radius?

A) 2

B) 3

C) 4

D) $\sqrt{5}$

Answer: C) 4

Method: $(x + 1)^2 + (y - 2)^2 = 1 + 4 + 11 = 16$, so $r = 4$.

Investigate further: Complete the square exactly: $x^2 + 2x = (x + 1)^2 - 1$ and $y^2 - 4y = (y - 2)^2 - 4$, so $(x + 1)^2 + (y - 2)^2 = 11 + 1 + 4 = 16$ and $r = 4$ ✓. With $g = 1, f = -2, c = -11$ the formula reads $r = \sqrt{g^2 + f^2 - c} = \sqrt{1 + 4 + 11}$. **Option D ($\sqrt{5}$) traps students who drop the constant 11 (leaving $r^2 = 1 + 4 = 5$) or flip the sign of c ; keep the sign fixed through the whole square.**

3. The point $(1, 2)$ lies on the circle $x^2 + y^2 = 5$. The equation of the tangent to the circle at $(1, 2)$ is?

A) $y = -\frac{1}{2}x + \frac{5}{2}$

B) $y = 2x$

C) $y = -2x + 5$

D) $y = \frac{1}{2}x$

Answer: A) $y = -\frac{1}{2}x + \frac{5}{2}$

Method: Radius gradient $= \frac{2}{1} = 2$; tangent gradient $= -\frac{1}{2}$; through $(1, 2)$: $y - 2 = -\frac{1}{2}(x - 1) \Rightarrow y = -\frac{1}{2}x + \frac{5}{2}$.

Investigate further: Option C uses the wrong sign ($-2x$ is the negative-reciprocal error recalling Day 1), B is the tangent being perpendicular to the perpendicular. **Split-variable test: tangent to $x^2 + y^2 = r^2$ at (x_1, y_1) is $xx_1 + yy_1 = r^2$: here $x + 2y = 5 \Rightarrow y = -\frac{1}{2}x + \frac{5}{2}$ ✓.**

4. A circle of form $x^2 + y^2 + 2gx + 2fy + c = 0$ passes through the origin. Deduce c .

- A) $c = 1$
 B) $c = 0$
 C) $c = g + f$
 D) $c = -1$

Answer: B) $c = 0$

Method: At the origin $(0, 0) \in$ circle means $0 + 0 + 0 + 0 + c = 0$ — the constant term is the value of the LHS at the origin, so $c = 0$.

Investigate further: Circle $x^2 + y^2 = 25$ has $c = -25$ and indeed does *not* pass through the origin. The constant records the origin's membership: $c = 0 \Leftrightarrow$ origin on the circle.

5. The perpendicular distance from the centre of $x^2 + y^2 = 25$ to the line $y = 2x + 3$ is?

- A) $\frac{3}{\sqrt{5}}$
 B) $\frac{3}{5}$
 C) 3
 D) 2

Answer: A) $\frac{3}{\sqrt{5}}$

Method: Line $2x - y + 3 = 0$; distance from $(0, 0) = \frac{|3|}{\sqrt{2^2 + (-1)^2}} = \frac{3}{\sqrt{5}}$.

Investigate further: Compare with $r = 5$: $\frac{3}{\sqrt{5}} < 5$, so the line cuts the circle. This is exactly the Day-1 formula put to the new line-circle test.

6. Which of the following is a tangent to the circle $x^2 + y^2 = 4$?

- A) $y = x + 2\sqrt{2}$
 B) $y = x + 4$
 C) $y = 2x + 2$
 D) $y = 2x + 4$

Answer: A) $y = x + 2\sqrt{2}$

Method: Line $x - y + 2\sqrt{2} = 0$ has distance from origin $\frac{2\sqrt{2}}{\sqrt{2}} = 2 = r$ — tangent. The others give $\frac{4}{\sqrt{2}} = 2\sqrt{2}$, $\frac{2}{\sqrt{5}}$, $\frac{4}{\sqrt{5}}$, none equal to 2.

Investigate further: Only option A places the origin exactly r away. **Check B:** $y = x + 4$ has distance $2\sqrt{2} > 2$, so it misses the circle. Shifting a tangent line a little breaks tangency —

| distance must *equal* r .

7. A chord of the circle $x^2 + y^2 = 13$ has length $4\sqrt{3}$. What is the perpendicular distance of the chord from the centre?

- A) 3
B) 2
C) $\sqrt{5}$
D) 1

Answer: D) 1

Method: Half-chord = $2\sqrt{3}$; $(2\sqrt{3})^2 = 13 - d^2 \Rightarrow 12 = 13 - d^2 \Rightarrow d^2 = 1 \Rightarrow d = 1$.

Investigate further: Fallback: remember $(\text{half-chord})^2 + (\perp \text{ dist})^2 = r^2$. Statement C ($\sqrt{5}$) is the trap of $13 - 8$ instead of $13 - 12$.

8. The line $y = mx + 4$ is a tangent to the circle $x^2 + y^2 = 4$. Find the two possible values of m .

- A) ± 1
B) $\pm\sqrt{3}$
C) ± 2
D) $\pm\sqrt{2}$

Answer: B) $\pm\sqrt{3}$

Method: Line $mx - y + 4 = 0$; tangency requires $\frac{|4|}{\sqrt{m^2+1}} = 2 \Rightarrow \sqrt{m^2+1} = 2 \Rightarrow m^2 = 3 \Rightarrow m = \pm\sqrt{3}$.

Investigate further: Both tangent lines pass through $(0, 4)$, symmetric about the y -axis — the two tangents from an external point. **How long is each tangent segment from $(0, 4)$ to its touch point?** $\sqrt{16 - 4} = \sqrt{12} = 2\sqrt{3}$ — a preview of Power of a Point (Day 5).

9. The circle with centre $(6, a)$ and radius 6 is tangent to the y -axis and passes through the point $(0, 8)$. Find a .

- A) 1
B) 3
C) 5
D) 8

Answer: D) $a = 8$

Method: Tangent to the y -axis ($x = 0$) since centre's x -distance $6 = r$. Through $(0, 8)$: $(0-6)^2 + (8-a)^2 = 36 \Rightarrow 36 + (8-a)^2 = 36 \Rightarrow a = 8$.

Investigate further: Setting $r = 6$ fixes the x -distance; passing through $(0, 8)$ forces the centre level with that point. **Option C (5) assumes the centre is $(6, a)$ with $(8-a) = 1$?** Check $(8-5)^2 = 9 \neq 0$.

10. Write down the length of the chord cut from the line $y = x + 1$ by the circle $x^2 + y^2 = 25$.

Answer: $7\sqrt{2}$

Method: Line $x - y + 1 = 0$; distance from the origin $= \frac{|1|}{\sqrt{2}} = \frac{1}{\sqrt{2}}$. Half-chord $= \sqrt{25 - \frac{1}{2}} = \sqrt{\frac{49}{2}} = \frac{7}{\sqrt{2}}$; full chord $= \frac{14}{\sqrt{2}} = 7\sqrt{2}$.

Investigate further: Note the line does not pass through the centre, so this is a genuine off-centre chord. If it were $y = x$ the chord would be a full diameter 10.

Section C Substitution & Structure

(≤ 90 s each)

Harder: hidden tangency, chord bisection, and the distance-vs-discriminant decision. No brute substitution.

1. A tangent to the circle $x^2 + y^2 = 144$ passes through $(20, 0)$ and meets the positive y -axis at T . What is the y -coordinate of T ? (after TMUA 2017 Paper 1 Q6)

- A) 12
B) 15
C) $\frac{49}{3}$
D) 20

Answer: B) 15

Method: Radius to touch point is $\perp$ tangent. Let $T = (0, t)$, line through $(20, 0)$ and $(0, t)$ has equation $x/20 + y/t = 1$, i.e. $tx + 20y - 20t = 0$. Distance from origin $= r = 12$: $|-20t|/\sqrt{t^2 + 400} = 12 \Rightarrow 400t^2 = 144(t^2 + 400) \Rightarrow 256t^2 = 57600 \Rightarrow t^2 = 225 \Rightarrow t = 15$ (positive). Check $t = 15$ gives line $3x + 4y = 60$, distance $60/5 = 12 \checkmark$.

Investigate further: The x -intercept 20 and $r = 12$ give a 5-12-13 scaled triple $(12, 16, 20)$. **If the external point were $(13, 0)$ instead, the y -int would be $12 \cdot 13/5$?** No, the triple changes.

2. The point $P(8, 6)$ is outside the circle $x^2 + y^2 = 36$. Tangents from P meet the circle at A and B . What is the length AB ? (adapted from TMUA 2018 Paper 1 Q7)

- A) $\frac{12\sqrt{13}}{13}$
B) $\frac{24\sqrt{13}}{13}$
C) $\frac{48}{5}$
D) $\frac{72}{5}$

Answer: C) $\frac{48}{5}$

Method: $OP = \sqrt{64 + 36} = 10$, $r = 6$, right triangle OAP has $AP = \sqrt{100 - 36} = 8$, distance from O to chord AB is $d = r^2/OP = 36/10 = 18/5$. Half-chord $= \sqrt{r^2 - d^2} = \sqrt{36 - 324/25} = 24/5$, so $AB = 48/5$.

Investigate further: The chord AB is the radical axis of the circle and the point-circle. **If P were at distance $2r$ from O , the chord length would be $r\sqrt{3}$ — check. Here $OP = 10 = 5r/3$, giving $48/5$.**

3. The circles $(x + 4)^2 + (y + 1)^2 = 64$ and $(x - 8)^2 + (y - 4)^2 = r^2$ have exactly one point in common. If $r < 13$, what is r ? (after TMUA 2019 Paper 1 Q6)

- A) 5
B) 8
C) 13
D) 21

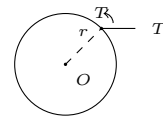
Answer: A) 5

Method: Centre distance $d = \sqrt{12^2 + 5^2} = 13$. One point $\Rightarrow$ internal tangency $r = |d - 8| = 5$ (since $r < 13$). External would give 21.

Investigate further: Two possibilities: $|r - 8| = 13$ gives $r = 5$ or 21; the condition $r < 13$ picks 5.
If $r > 13$, the other is 21.

4. A circle has centre $(9, 11)$ and radius 5. A regular hexagon $ABCDEF$ is inscribed with all vertices on the circle. What is the area of the hexagon? (adapted from TMUA 2017 Paper 1 Q9)

- A) $18\sqrt{3}$
B) $36\sqrt{3}$
C) $\frac{75\sqrt{3}}{2}$
D) 75



Answer: C) $\frac{75\sqrt{3}}{2}$

Method: Regular hexagon = 6 equilateral triangles side $r = 5$: area = $6 \cdot \frac{\sqrt{3}}{4} r^2 = \frac{3\sqrt{3}}{2} r^2 = \frac{75\sqrt{3}}{2}$.

Investigate further: Centre $(9, 11)$ is irrelevant — translation does not change area. **If it were a square, area would be $2r^2 = 50$.**

5. The line $y = 2x + 5$ is a tangent to the circle $x^2 + y^2 = r^2$. Which r^2 makes the distance from the centre to the line equal to r ?

- A) 1
B) 5
C) 20
D) 25

Answer: B) 5

Method: Distance = $|5|/\sqrt{5} = \sqrt{5} = r$, so $r^2 = 5$.

Investigate further: Tangent to centred circle $r = |c|/\sqrt{a^2 + b^2}$. **If line were $y = 2x + 10$, r^2 would be 20.**

6. The circle $x^2 + y^2 - 4x + 2y - 11 = 0$ and the line $x - y + 1 = 0$: determine the number of intersection points without solving the system. (after MAT 2019 Q6)

- A) 0 (line misses)

- B) 1 (tangent)
 C) 2 (secant)
 D) Infinitely many (line is diameter)

Answer: C) 2 (secant)

Method: Centre $(2, -1)$, $r = 4$, distance to $x - y + 1 = 0$ is $|2 + 1 + 1|/\sqrt{2} = 4/\sqrt{2} = 2\sqrt{2} < r$, so 2 points.

Investigate further: Compare d vs r , never substitute. **Diameter would need $d = 0$.**

7.

The circle $x^2 + y^2 - 8x - 4y - 21 = 0$ ($r = \sqrt{41}$) is cut by the y -axis ($x = 0$). What is the length of the chord intercepted on the y -axis?



- A) $2\sqrt{21}$
 B) $2\sqrt{41}$
 C) 8
 D) 10

Answer: D) 10

Method: Centre $(4, 2)$, distance to $x = 0$ is 4, half-chord $\sqrt{41 - 16} = 5$, length 10. Also $x = 0 \Rightarrow y^2 - 4y - 21 = 0 \Rightarrow y = 7, -3$, distance 10.

Investigate further: Two methods: distance vs. direct substitution give 10. **If the line were $x = 4$ (through centre), chord would be diameter $2\sqrt{41}$.**

8. The line $y = kx + 5$ is tangent to the circle $x^2 + y^2 - 6x - 8y + 24 = 0$ (centre $(3, 4)$, $r = 1$). How many distinct k satisfy this? (after TMUA 2020 Paper 1 Q7)

- A) 0
 B) 1
 C) 2
 D) Infinitely many

Answer: C) 2

Method: Point $(0, 5)$ is distance $\sqrt{10} > 1$ from centre, so external: two tangents. Solve $|3k - 4 + 5|/\sqrt{k^2 + 1} = 1$ gives 2 slopes.

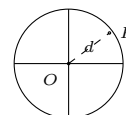
Investigate further: From an external point exactly two tangents; from a point on the circle one; from inside zero. **Here $(0, 5)$ is $\sqrt{10}$ away, so two.**

Section D Challenge

(TMUA / SMC / BMO1 difficulty)

Hardest 5: distance invariants, hidden right angles, and a construction you must spot.

1. The circles $C_1 : (x+2)^2 + (y-3)^2 = 18$ and $C_2 : (x-7)^2 + (y+6)^2 = 2$ have centres O_1, O_2 . What is the *shortest* distance between a point on C_1 and a point on C_2 ? (adapted from TMUA 2018 Paper 1 Q7 / 2022 Paper 1 Q3)



- A) $5\sqrt{2} - 4$
 B) $5\sqrt{2} - 5$
 C) $5\sqrt{2}$
 D) $5\sqrt{2} + 5$

Answer: C) $5\sqrt{2}$

Method: $O_1 = (-2, 3)$, $O_2 = (7, -6)$, $O_1O_2 = \sqrt{9^2 + 9^2} = 9\sqrt{2}$, $r_1 = 3\sqrt{2}$, $r_2 = \sqrt{2}$, sum $4\sqrt{2}$, shortest $= 9\sqrt{2} - 4\sqrt{2} = 5\sqrt{2}$. Externally separated ($d > r_1 + r_2$).

Investigate further: Shortest = centre distance minus sum of radii (external). If one circle contained the other, it would be $|r_1 - r_2| = d$. For these numbers, $5\sqrt{2} \approx 7.07$.

2. The segment joining $(3, 3)$ and $(7, 5)$ is a diameter of circle C . C is translated 3 left, reflected in the x -axis, then enlarged scale factor 4 about its centre. What is the equation of the final circle? (after TMUA Specimen Paper 1 Q9)

- A) $(x - 2)^2 + (y - 4)^2 = 80$
 B) $(x - 2)^2 + (y + 4)^2 = 80$
 C) $(x - 2)^2 + (y - 4)^2 = 320$
 D) $(x - 2)^2 + (y + 4)^2 = 320$

Answer: B) $(x - 2)^2 + (y + 4)^2 = 80$

Method: Original centre $(5, 4)$, $r^2 = (2)^2 + 1^2 = 5$. Translate 3 left: $(2, 4)$, reflect in x -axis: $(2, -4)$, enlarge $4\times$ about centre: $r^2 = 5 \cdot 16 = 80$, so $(x - 2)^2 + (y + 4)^2 = 80$.

Investigate further: Enlargement about the centre leaves the centre fixed and scales radius by 4. The 320 options would be $r^2 \cdot 4$ too large.

3. Circle O radius 6, $\angle POQ \geq \pi/2$, area $[POQ] = 9\sqrt{3}$. For variable R on the major arc PQ , what is the *greatest possible* area of $\triangle PRQ$? (adapted from TMUA 2022 Paper 1 Q14)



- A) $18 + 9\sqrt{3}$
 B) $27\sqrt{3}$
 C) $27 + 9\sqrt{3}$
 D) $36 + 9\sqrt{3}$

Answer: B) $27\sqrt{3}$

Method: $[POQ] = \frac{1}{2}r^2 \sin \theta = 18 \sin \theta = 9\sqrt{3} \Rightarrow \sin \theta = \sqrt{3}/2$, $\theta = 120^\circ$. Chord $PQ = 6\sqrt{3}$, maximal height 9, area $27\sqrt{3}$.

Investigate further: If θ had been 60° , max would be $9 + 18 = 27$.

4. Let $P(p, q)$ and circle $C : x^2 + 2fx + y^2 + 2gy + h = 0$ with centre $C_0 = (-f, -g)$ and $L = |PC_0|$. Which set is *minimal sufficient* to compute L ? (after MAT 2020 Q1H)

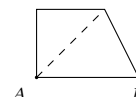
- A) f, g, h
- B) f, g, p, q
- C) f, h, p, q
- D) g, h, p, q

Answer: B) f, g, p, q

Method: $L^2 = (p + f)^2 + (q + g)^2$ needs f, g, p, q ; h irrelevant.

Investigate further: Tangent length needs h , not L .

5. $ABCD$ is a cyclic quadrilateral with perpendicular diagonals intersecting at E . The line through E perpendicular to AB meets CD at F . Which statement is *necessarily true*? (adapted from BMO1 2010 Q4)



- A) F is the midpoint of CD (Brahmagupta)
- B) F bisects $\angle CED$
- C) $EF = AB/2$
- D) $ABCD$ must be an isosceles trapezium

Answer: A) F is the midpoint of CD (Brahmagupta)

Method: Brahmagupta: In a cyclic quadrilateral with $\perp$ diagonals, the perpendicular from the intersection to a side bisects the opposite side.

Investigate further: The other options fail for a general kite.

Top 5 Patterns — Worksheet #2

- Complete the square to read centre and radius.** In $x^2 + y^2 + 2gx + 2fy + c = 0$ the centre is $(-g, -f)$ and $r = \sqrt{g^2 + f^2 - c}$. Strip the linear terms into squares before touching c . (see A1, A2, A4, B2, C8)
- Tangency = distance-to-line equals radius.** For a tangent, $\frac{|ax_0 + by_0 + c|}{\sqrt{a^2 + b^2}} = r$. Never substitute line into circle. (see A9, B6, C1, D4)
- Tangent at a point: perpendicular to the radius / split variables.** On $x^2 + y^2 = r^2$, tangent at (x_1, y_1) is $xx_1 + yy_1 = r^2$. (see A7, B3, C4, D3)
- Chord: half-chord = $\sqrt{r^2 - d^2}$.** The perpendicular from the centre bisects the chord and makes a right triangle with half-chord, distance, and radius. (see A10, B7, B10, C5, D1)
- Line-circle positioning by comparing d to r .** $d < r$ secant, $d = r$ tangent, $d > r$ external. (see B5, C3, C8)

Common Traps — Worksheet #2

- Forgetting to halve the linear coefficients.** $x^2 + y^2 - 6x - 8y + 24 = 0$ has centre $(3, 4)$, not $(-6, -8)$. Centre is $(-g, -f)$ where $2g, 2f$ are the linear terms. (see A1, A4, B2)
- Negative reciprocal error on the tangent slope.** Tangent to a circle is perpendicular to the radius, not a mere negation or inversion. (see A6, A7, B3)

- 3. Applying tangency when the point is not on the circle.** The tangent-at-a-point formula only holds for points on the circle; check $x_1^2 + y_1^2 = r^2$ first. *(see B3, C4, D3)*
- 4. Generic lines through the centre are diameters, not tangents.** A line through the centre always cuts twice; tangency needs the perpendicular distance to be exactly r , not 0. *(see B5, C3, C8)*
- 5. Mixing up distance-to-centre with chord length.** The perpendicular distance d is one leg; the half-chord is the other. Squaring and subtracting, then doubling. *(see B7, B10, C5, D1)*

“Tangency is a distance statement in disguise: compare d to r , never solve the intersection.”

Found a mistake or want to contribute a sheet?
Visit the open-source project at github.com/The-CerealDev/speed-maths